§3.2 Lyapunov functions

We can prove important results about the stability of fixed points of a given flow $\phi_t(\bar{x})$ if we can find some functions that behave in a certain way under the flow.

Definition A continuously differentiable (C^1) function $V(\bar{x}): R^n \to R$ is a Lyapunov function for a flow $\phi_t(\bar{x})$ on a domain D containing the origin (fixed point) if

(i) $V(\bar{0}) = 0$ and $V(\bar{x}) > 0$ for $\bar{x} \neq \bar{0}$ in D and

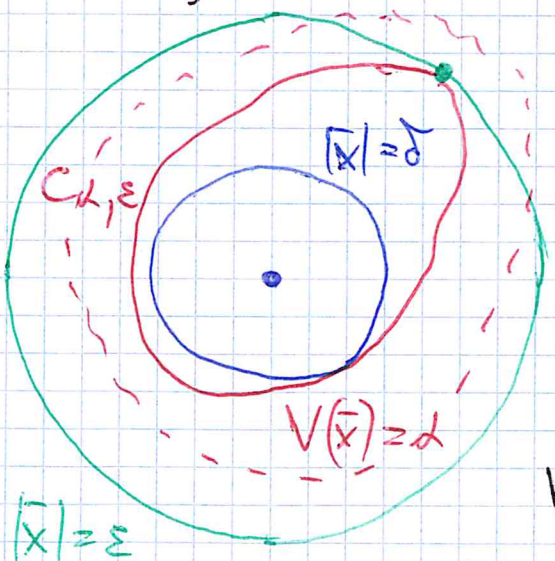
(ii) $\dot{V} = \bar{f} \cdot \nabla V \leq 0$ in D (or $V(\phi_t(\bar{x})) \leq V(\bar{x})$ for $\forall t \geq 0$).

Then we have these 2 theorems.

Theorem Lyapunov's 1st theorem

Suppose that a dynamical system $\dot{\bar{x}} = \bar{f}(\bar{x})$ has a fixed point at the origin. If a Lyapunov function exists, then the origin is Lyapunov stable.

Proof Let's take small enough $\varepsilon > 0$ such that $\{|\bar{x}| < \varepsilon\} \subset D$. Let $\alpha = \min\left(V(\bar{x}), |\bar{x}| = \varepsilon\right) > 0$ ($V(\bar{x}) = 0$ only at the origin!). Now choose δ: $\{|\bar{x}| < \delta\} \subset C_{\alpha,\varepsilon} = \{\bar{x} : V(\bar{x}) < \alpha, |\bar{x}| < \varepsilon\}$, then $\{|\bar{x}| < \delta\} \subset C_{\alpha,\varepsilon} \subseteq \{|\bar{x}| < \varepsilon\}$.



Then if $\bar{x} \in \{|\bar{x}| < \delta\}$, we have $V(\phi_t(\bar{x})) \leq V(\bar{x}) < \alpha$, $\Rightarrow \underline{\phi_t(\bar{x}) \in \{|\bar{x}| < \varepsilon\}}$.

Thus, for $\forall \varepsilon > 0$ found $\delta > 0$: $\phi_t(\bar{x}) \in N_\varepsilon(\bar{0})$ if $\bar{x} \in N_\delta(\bar{0})$.

In other words, the definition of Lyapunov stability is fulfilled.

Theorem Lyapunov's 2nd theorem

If in addition $\dot{V} < 0$ for $\bar{x} \neq \bar{0}$ then V is a strict Lyapunov function and the origin is asymptotically stable.

Proof From 1st theorem we know that starting in D, $\phi_t(\bar{x})$ stays in D. So for any $\bar{x}_0 \in D$ $V(\bar{x}_0) > 0$ and $V(\phi_t(\bar{x}_0))$ is decreasing as t increases. Thus, V must tend to some $V_0 \geq 0$. Suppose $V_0 > 0$, then $V(\phi_t(\bar{x}_0))$ is bounded away from 0 and $\phi_t(\bar{x})$ is also bounded away from $\bar{0}$. If $|\phi_t(\bar{x}_0)| \neq 0$, then $\dot{V}(\phi_t(\bar{x}_0)) < 0$ is also bounded away from 0, so $\dot{V}(\phi_t(\bar{x}_0)) < -b < 0$ for some $b > 0$. But then

$$V(\phi_t(\bar{x}_0)) = V(\bar{x}_0) + \int_0^t \dot{V}(\phi_t(\bar{x}_0))\,dt \leq V(\bar{x}_0) - bt$$

and the right-hand side eventually becomes negative for large enough t, which is impossible since $V(\bar{x}) \geq 0$ for $\bar{x} \in D$.

Thus, $V_0 = 0$ and $\phi_t(\bar{x}_0) \to \bar{0}$ for any $\bar{x}_0 \in D$ since $V(\bar{x}) = 0$ is possible only if $\bar{x} = \bar{0}$. Thus, asymptotically stable.

Example Analize stability of the origin $\bar{x} = \bar{0}$ for

$$\begin{pmatrix} \dot{x} \\ \dot{y} \end{pmatrix} = \begin{pmatrix} -2x \\ x - y \end{pmatrix}, \quad \hat{A} = \begin{pmatrix} -2 & 0 \\ 1 & -1 \end{pmatrix}$$

This is a linear system, Jacobian $\Rightarrow \lambda_1 = -1$, $\lambda_2 = -2$ and is a stable node.

Let's illustrate stability using Lyapunov function.

Try $V(x,y) = ax^2 + by^2$, $a,b>0$, clearly $V \geq 0$ and $V=0$ only at $\bar{x}=\bar{0}$. Compute

$$\dot{V} = V_x \dot{x} + V_y \dot{y} = 2ax(-2x) + 2by(x-y) = -2b\left(\frac{2a}{b}x^2 - xy + y^2\right)$$

(i) Take $b = 8a$, $\Rightarrow \dot{V} = -2b\left(\frac{x}{2} - y\right)^2 \leq 0$, so Lyapunov but not strict as $\dot{V}=0$ on $\{y = \frac{x}{2}\}$, which extends beyond the origin $\bar{x}=0$. Still, can claim Lyapunov stability (LS) using L1.

(ii) Take $b = 2a$, $\Rightarrow \dot{V} = -2b\left[\frac{3}{4}x^2 + \left(\frac{x}{2} - y\right)^2\right] < 0$ for any $\bar{x} \neq \bar{0}$, $\Rightarrow$ strict Lyapunov function is found, so by L2 origin is asymptotically stable (AS).

Even when L2 conditions are not fulfilled, can sometimes demonstrate stability of the origin.

Theorem La Salle's Invariance Principle

If V is a Lyapunov function on a compact (closed and bounded) domain D, then $\forall \bar{x}$: $\phi_t(\bar{x})$ is bounded $\exists c$ such that ω-limit set of $\bar{x}$

$$\omega(\bar{x}) \subseteq M_c = \{\bar{z} : \dot{V}(\phi_t(\bar{z})) = c \ \forall t \geq 0\}.$$

Or, equivalently, $\phi_t(\bar{x})$ tends to an invariant subset of a set $\{\bar{z} : \dot{V}(z) = 0\}$.

Proof Choose $\bar{x} \in D$ and let $c = \inf_{t \geq 0} V(\phi_t(\bar{x}))$. Since D is compact and $\phi_t(\bar{x})$ is bounded, ω-limit set exists. Take $y \in \omega(\bar{x})$, $\Rightarrow$ by definition, $\exists$ a sequence $t_1, t_2, \ldots$ such that $\phi_{t_n}(\bar{x}) \to \bar{y}$ for $n \to \infty$. As V is continuous and $\dot{V} \leq 0$ on orbits, find that $V(\bar{y}) = c$.

Now show that $V(\phi_t(\bar{y})) = c$. Suppose not

and $V(\phi_T(\bar{y})) < c$ for some T. But we know that $\phi_{t_n}(\bar{x}) \to \bar{y}$ as $n \to \infty$, so

$$V(\phi_{t_n+T}(\bar{x})) = V(\phi_T(\phi_{t_n}(\bar{x}))) \to V(\phi_T(\bar{y})) < c.$$

But this means that there exists time $\tau = T + t_n$ such that $V(\phi_\tau(\bar{x})) < c$, which contradicts the definition of c! Thus, $V(\phi_t(\bar{y})) = c$. In other words, $\bar{y}$ is a part of an invariant subset of M_c, which is the same as $\{\bar{z} : \dot{V}(\bar{z}) = 0\} \cap D$.

Note:

(*) $\omega(\bar{x})$ doesn't need to be a fixed point in general, so a more general statement than L1 or L2.

But in practice we'll be using La Salle to determine stability of fixed points (more on this later).

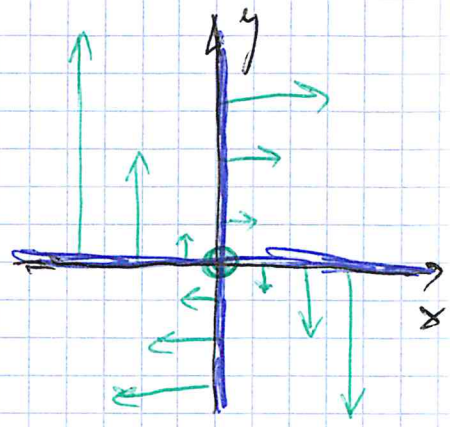
(**) If the only invariant subset of $\{\bar{z} : \dot{V}(\bar{z}) = 0\}$ is $\bar{x} = \bar{0}$, then the origin is asymptotically stable by La Salle. Thus, a slightly more stringent set of conditions results in a ~~more~~ stronger statement than L1, and don't need Lyapunov function to be strict!

Example Explore stability of the origin for $\begin{pmatrix} \dot{x} \\ \dot{y} \end{pmatrix} = \begin{pmatrix} y - xy^2 \\ -x^3 \end{pmatrix}$

Note that without $-xy^2$ term this system would be Hamiltonian with $H(x,y) = \frac{1}{2}y^2 + \frac{1}{4}x^4$. Take $V(x,y) = H(x,y)$, then

$\dot{V} = H_x\dot{x} + H_y\dot{y} = x^3(y - xy^2) + y(-x^3) = -x^4y^2 \leq 0$, $\Rightarrow$ H is indeed a Lyapunov function in R^2.

It is not a strict one, as $\dot{V} = 0$ on $\{x=0$ and/or $y=0\}$, i.e. also outside the origin, so can't use L2. But all members of the set $\{\bar{z} : \dot{V}(\bar{z}) = 0\}$ with $x \neq 0$ or $y \neq 0$ are not invariant (see figure, examine the system), thus only $\bar{x} = \bar{0}$ is the invariant subset of $\{\bar{z} : \dot{V}(\bar{z}) = 0\}$. Thus, by La Salle, origin is AS.



We can use Lyapunov & La Salle's theorems to establish basin of attraction (domain of stability, DoS) of a fixed point. Here is an Algorithm:

(1) Choose a function $V(\bar{x}): R^n \to R$ and domain D such that (i) $V(\bar{x}) \geq 0$ in D and $V(\bar{x}) = 0$ only for $\bar{x} = 0$
(ii) $\dot{V}(\bar{x}) \leq 0$ in D (this might limit D!)

(2) Find k such that $C_k = \{\bar{x} : V(\bar{x}) < k\} \subseteq D$

(3) Adjust k (or V) such that the only invariant subset of $C_k \cap \{\bar{z} : \dot{V}(\bar{z}) = 0\}$ is the origin. Then La Salle gives that $C_k \subseteq DoS(\bar{0})$

(4) Take different V to get another C_k. The union of all C_k forms a better (larger) estimate of $DoS(\bar{0})$.